

In Next.js, the purpose of `NextResponse.rewrite(new URL('/api/final-endpoint', request.url))` is to map an incoming request to a different destination path **internally**, acting as a [URL proxy](#). 

Its key functionalities and characteristics include:

- **URL Masking:** It masks the destination path, meaning the content from `/api/final-endpoint` is served, but the URL in the browser's address bar remains unchanged for the user.
- **Internal Routing:** Unlike a redirect, which sends a 3xx status code and forces the browser to make a new request to a new URL, a rewrite handles the routing entirely on the [server-side](#).
- **Primary Use in Middleware:** It is most commonly used within [Next.js Middleware](#) to intercept requests and route them to specific pages or API endpoints based on logic (e.g., authentication, A/B testing, or geolocation) without the user noticing.
- **Request Forwarding:** It allows you to proxy requests to external services or internal routes while keeping your application's public URL structure clean.
- **Limitation in Route Handlers:** While useful in Middleware, using `NextResponse.rewrite()` directly inside [App Router Route Handlers](#) (e.g., `route.ts`) is currently not supported.